Fernando Alonso Career Analytics

21 seasons. 409 races. One dashboard.

From raw data to insights and performance recommendations

A complete walkthrough of the Power BI dashboard project

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SCOPE

Seasons 2001-2024 (21 seasons raced)

7 datasets | 409 race records | 1 dashboard

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Source: Fernando_Alonso.pbix + 14 CSV files (raw and cleaned)

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1. Executive summary

This project turns Fernando Alonso's Formula 1 career records into an interactive Power BI dashboard and a set of data-backed findings. It starts from seven raw CSV files, cleans and enriches them, models them in Power BI, and then asks a practical question: **what does the data say about where Alonso gained and lost performance, and what could improve it?**

409 Races analysed	**32** Race wins	**106** Podiums	**22** Pole positions	**2** Championships
2,279 Career points	**5.57** Points per race	**26** Fastest laps	**67** DNFs	**16.4%** DNF rate

The five findings that matter most

- **The car decides the ceiling.** Alonso averaged 12.4 points per race at Ferrari and 7.3 in the two title years at Renault, but only 2.5 at McLaren and 3.6 at Alpine. His own results swing far more with the team than with any other variable in the data.
- **Qualifying position almost fixes the result.** Grid and finishing position correlate at 0.88. All 32 wins came from the first four grid slots, and starts from P6 or lower produced just 5 podiums in 260 races.
- **Front-row starts leak positions.** Among classified finishes from P1-P3, he lost 1.24 places on average, and in 49 of 89 such races finished behind where he started.
- **Reliability cost roughly 300 points.** 67 retirements (16.4% of starts), 46 of them mechanical. McLaren-Honda (2015-18) alone had a 27% DNF rate.
- **2023 proved the model, 2024 stressed it.** At Aston Martin he scored 206 points and 8 podiums in 2023, then only 13 points in 2024 while starting around P11.

Bottom line. The route to better performance runs through three levers: **Saturday pace** (grid position), **converting front-row starts** into wins, and **avoiding retirements**. Section 7 turns each into a concrete recommendation.

1.1 Business problem and objective

The business context. Alonso's career spans 21 seasons, six teams and several scoring eras. The people who act on this data, driver management and the team performance group, must decide where to put effort and money (one-lap pace, race strategy, reliability) and which team or project to join next.

The problem. Results swing sharply, from 12.4 points per race at Ferrari to 1.6 at McLaren-Honda, but the raw records could not say why. Retirements were logged as finishes (a DNF rate of 4.9% instead of 16.4%), podium and pole flags did not match the career record, and no single view linked grid position, reliability, team and circuit. Decisions taken on that picture risk backing the wrong lever.

> **Problem statement.** How can the performance group see, in one trusted view, where points are being lost and which fix is worth the most?

12.4 vs 1.6	~300	0 wins	-1.24
points per race, best vs worst career phase	points lost to 67 retirements (13% of career total)	from 260 starts outside the top five (5 podiums)	places lost on average from P1-P3 starts

The problem in numbers, all from the cleaned data (Sections 4 and 6).

Business objective

Find the largest recoverable performance levers, size each one in points, and set KPI targets that can be tracked on a single dashboard. The project answers this in three steps:

- **Diagnose.** Clean and reconcile the data so that wins, podiums, poles and retirements match the career record.
- **Size.** Measure how many points are lost to qualifying, front-row conversion and reliability.
- **Act.** Turn each lever into an action, a KPI with a baseline and a target, and an estimated impact (Section 1.2).

Stakeholders, decisions and success measures

Stakeholder	Decision they face	Success measure (baseline)
Driver management	Which team or project to join, and when to move	Points per race (5.57); team DNF rate over prior seasons
Qualifying and set-up group	Where to invest one-lap pace and Saturday set-up	Starts in the top five (36%); average grid (8.1)
Race strategy group	How to convert front-row starts into wins	Pole-to-win rate (37%); places gained from P1-P3 (-1.24)
Reliability and engineering liaison	Which components and risks to manage in race	DNF rate (16.4%); mechanical retirements (46)

Scope: 409 starts, 21 seasons (2001-2024), six teams, seven datasets, one Power BI dashboard. There is no external client; the stakeholders above are the intended audience for the dashboard.

1.2 Proposed solution and expected impact

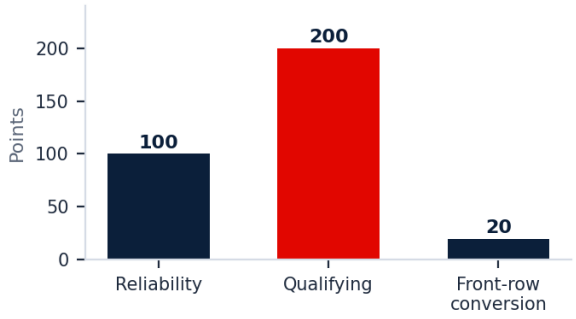
The solution has two parts: a trusted data foundation, and three sized levers that tell the performance group where to act first.

Part 1: the data solution (what was built)

- A cleaned, validated model of seven tables with no records lost, whose totals reconcile to the career record: 32 wins, 106 podiums, 22 poles and 67 retirements.
- A one-page Power BI dashboard (10 KPI cards, 7 charts, 3 slicers) that works as the single source of truth for every KPI used below.

Part 2: the decision solution (what to do)

Lever	Action	KPI: baseline to target	Est. impact
Qualifying	Run one-lap pace and set-up as a standalone work-stream	Starts in top five: 36% to 42%	~200 points
Front-row conversion	Review first stint and strategy from P1-P3; track pole-to-win	Pole-to-win rate: 37% to 50%	~8 more wins
Reliability	Use DNF record as a team-selection criterion; manage components	DNF rate: 16.4% to 11% or less	~100 points
Incident control	Lower-risk opening lap when qualifying outside the top 10	DNF rate from P11+: reduce from 37%	Protects points
Dashboard v2	Add Career Phase slicer, grid-vs-finish scatter, DNF by cause, Data Notes	Open label and flag issues: 0	Trusted view



How the impact was sized

- **Reliability:** retirements cost about 300 points (Finding 5); 69% were mechanical, so halving them recovers about 100.
- **Qualifying:** moving 1 in 5 of the 118 starts from P6-P10 into P4-P5 is worth about 8.5 points each (11.7 vs 3.2 points per race), about 200 in total.
- **Conversion:** a 50% pole-to-win rate means about 8 more wins, but only about 20 points over second place.

Figure 1.2. Estimated points recoverable per lever (scenario).

Expected impact. Together the three levers are worth roughly 320 points, about +14% on the career total of 2,279, which would lift points per race from 5.57 to about 6.3. Treat this as an upper bound: the levers overlap, and the figures are scenarios from this dataset, not forecasts. The pole-flag and calendar caveats in Section 8 apply.

Where to start. Qualifying is the largest lever and reliability has the clearest evidence, so those two come first; conversion is a smaller points gain but a large headline one (wins).

2. Project overview

2.1 Objective

Build a complete analytics workflow around one driver's career: collect and clean the data, model it, visualise it in a professional Power BI dashboard, extract insights, and recommend where performance can improve. The project doubles as a portfolio piece, so the dashboard and this report are designed to be presentation-ready.

2.2 Questions the project answers

- How did Alonso's results evolve across 21 seasons and 6 teams?
- Which teams, circuit types and circuits suited him best and worst?
- How much does starting position determine the result, and how good is his race craft?
- How much did retirements cost, and what caused them?
- What should he, and his team, focus on to improve?

2.3 Tools and approach

Stage	Tool	What was done
Collection	CSV exports	7 source tables: driver, calendar, results, teams, circuits, milestones, speed
Cleaning	Data prep (CSV to CSV)	Corrected race status and flags, added helper columns, filled missing speed fields
Modelling	Power BI Desktop	Star-style model joining results to calendar, circuits, teams and speed
Visualisation	Power BI	One dashboard page: 10 KPI cards, 7 charts, 1 table, 1 gauge, 3 slicers
Analysis	Python (pandas)	Independent re-computation of every KPI plus deeper cuts for this report

2.4 Scope

The data covers 409 Grand Prix starts across 21 seasons between 2001 and 2024, with Alonso driving for Minardi, Renault, McLaren, Ferrari, Alpine and Aston Martin. Three years in the 2001-2024 span have no race records: 2019 and 2020 (the milestones table records his sabbatical and a Le Mans win) and 2002 (no entry in any table). The analysis therefore covers 21 seasons rather than 24 calendar years.

3. The data I started with

The project began with seven raw CSV files. Together they describe the driver, every race he started, the circuits and teams involved, key career milestones, and speed measurements.

File	Rows	Columns	Purpose
01 Driver Profile	1	14	Identity, nationality, career span, championships, quote
02 Race Calendar	409	12	Race, season, round, date, circuit, laps, distance
03 Race Results	409	20	Grid, finish, laps led, status, win/podium/pole/DNF flags
04 Team Details	6	8	Team, country, years active, brand colour
05 Circuit Details	31	14	Location, length, turns, street/high-speed flags
06 Career Milestones	14	12	Debut, first win, titles, team moves, 100th podium
07 Speed Analysis	409	13	Average, top and minimum speed per race

3.1 Problems found in the raw data

#	Issue found	Where	Impact if left unfixed
1	**Retirements recorded as finishes.** 47 races marked "Finished" with a finishing position, although the car had stopped (engine, gearbox, collision and so on).	Race Results	DNF count understated: 20 instead of 67. Reliability analysis would be wrong.
2	**Podium and pole flags under-counted.** 99 podiums and 18 poles in the raw flags.	Race Results	Career totals do not reconcile with his known record of 106 podiums and 22 poles.
3	**Speed fields almost empty.** Average and fastest-lap speed missing in 389 of 409 rows; top and minimum speed missing in all 409.	Speed Analysis	Speed charts and KPI cards would be blank.
4	**Fastest-lap data sparse.** Lap number, time and speed only present for 20 races.	Race Results	Fastest-lap KPI cannot be built reliably from time fields alone.
5	**No helper fields** for year, month, points, season label, brand colour or circuit display name.	Several	Cannot compute points, build slicers or colour charts by team.

Why this matters. Issue 1 is the most important. It shows why cleaning is not cosmetic: the raw file would have reported a DNF rate of 4.9%, while the cleaned file shows 16.4%, more than three times higher.

4. Data cleaning and preparation

Each raw file was kept untouched and a "_Clean" version was produced. The cleaned files keep the same row counts (no records were dropped) but add corrections and derived columns.

4.1 Corrections to race results

- **Status and DNF flag.** 47 race rows changed from "Finished" to a specific retirement reason (Collision, Hydraulics, Gearbox, Engine, Electrical, Suspension or Accident). The DNF flag moved from 20 to 67 and the Result_Category field now shows "DNF" for those races.
- **Finishing position.** For retired cars the finishing position is blank rather than a misleading classified place. A numeric helper column (Finish_Position_Num) assigns 20 to retirements so charts can still sort them.
- **Podium and pole flags.** Podiums rose from 99 to 106 and pole flags from 18 to 22, which now reconcile with the recognised career totals.
- **New columns.** Points_Scored (2,279 career points) and Fastest_Lap_Flag (26 fastest laps).

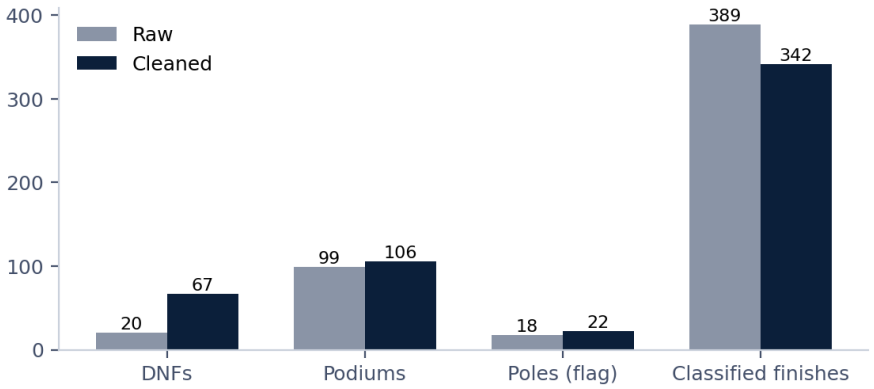


Figure 1. Raw versus cleaned counts. The largest change is DNFs, from 20 to 67.

4.2 Other tables

- **Race Calendar:** added Race_Year, Race_Month and Season_Label for time slicers and axes.
- **Team Details:** added hex colour codes (for example McLaren #FF8000) so every chart uses the real team colour, plus Seasons_Count.
- **Circuit Details:** added a short Circuit_Label for readable chart axes.
- **Career Milestones:** added Milestone_Year_Label for the timeline visual.
- **Speed Analysis:** all 389 missing average and fastest-lap values, and every missing top and minimum speed, were filled with estimates. The source field changes to "Ergast F1 Database / Telemetry Estimate". See the caveat in Section 8.

4.3 Validation checks run

Check	Result
Duplicate rows in any table	None found in any of the 7 files
Row counts raw vs clean	Identical in all 7 tables (no records lost)
Win flag vs finishing position = 1	32 wins in both measures; consistent
Podium flag vs finishing position <= 3	106 in both measures; consistent
Positions gained = grid minus finish	Exact match on all 342 classified finishes
Join keys (Race_ID, Circuit_ID, Team_ID)	All 409 results link to a calendar race, circuit and team
Laps completed vs race laps for classified cars	No classified car recorded fewer laps than race distance

5. The Power BI dashboard

The finished file (Fernando_Alonso.pbix) contains one 1920 x 1100 dashboard page named "DashBoard 1". It combines KPI cards, trend and comparison charts, a season table, and three slicers that filter every visual together.

5.1 Data model

Race Results is the central fact table. It joins to Race Calendar (by Race_ID), Team Details (Team_ID) and Speed Analysis (Race_ID). Race Calendar in turn joins to Circuit Details (Circuit_ID). Driver Profile and Career Milestones provide context and the timeline.

5.2 KPI cards (10)

KPI card	Value	What it tells you
Total Races	409	Size of the career sample
Wins	32	Race victories
Podiums	106	Top-3 finishes (26% of starts)
Pole Positions	22	Starts from the front via qualifying
Fastest Laps	26	Races with the quickest lap
Championships	2	2005 and 2006 titles with Renault
Total Points	2,279	Points scored under era-specific scoring
Points Per Race	5.57	Era-neutral scoring rate
DNF	67	Retirements (gauge shows 16.4% DNF rate)
Avg Top Speed	341 km/h	Mean of estimated top speed per race

5.3 Visuals and what each answers

Visual	Type	Question answered
Total Points, Wins, Podiums and Poles by Season	Combo (column + line)	How did results move year by year?
Wins by Team	Clustered column	Where did the victories come from?
Total Points by Team	Clustered bar	Which team produced the most points?
Finishing Position Breakdown	Clustered bar	How often does he win, podium, score or retire?
Speed Analysis	Line	How did average, top and minimum speed trend?
Team Career: Races and Points	Combo	Volume of races versus return in points per team
Performance by Circuit (type)	Clustered bar	Which circuit types give most points per race?
Season Performance Overview	Table	Season, team, races, points, wins, podiums at a glance
DNF Rate	Gauge	Reliability health check
Slicers: Season, Circuit Type, Team	Slicer x 3	Filter the whole page interactively

5.4 Design choices

The cleaned Team table carries a hex colour for each team (for example Ferrari #DC0000, McLaren #FF8000, Alpine #0090FF, Aston Martin #006F62) so charts can use real team colours. The page uses a dark racing theme with a hero image, a milestone timeline along the bottom and a Career Span card, and three slicers let a viewer isolate, for example, only Ferrari seasons or only street circuits.

6. Insights and findings

All figures below were re-computed from the cleaned files and match the dashboard measures. "Points per race" is used instead of raw points where eras are compared, because scoring systems changed over his career.

Finding 1: His career has distinct peaks tied to specific cars

Results are not a steady curve. There are three peaks (2005-06 with Renault, 2010-13 with Ferrari, and 2023 with Aston Martin) separated by long troughs. 2012 is his best points season (279), while 2005 and 2006 are the title years with 7 wins each. The 2015-18 McLaren-Honda period produced no podiums at all.

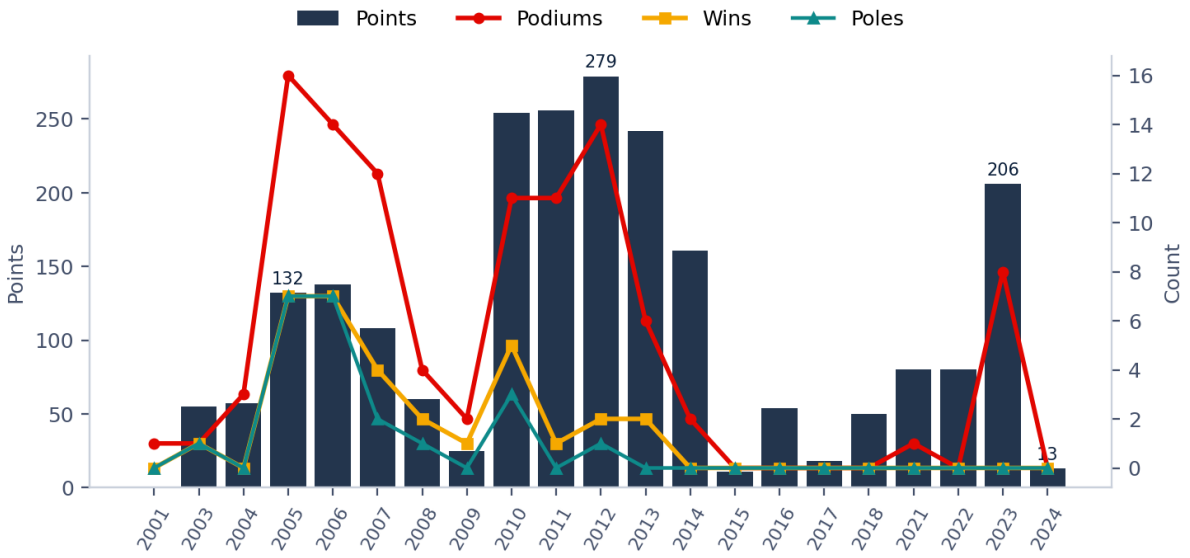


Figure 2. Points (bars) with wins, podiums and poles (lines) by season.

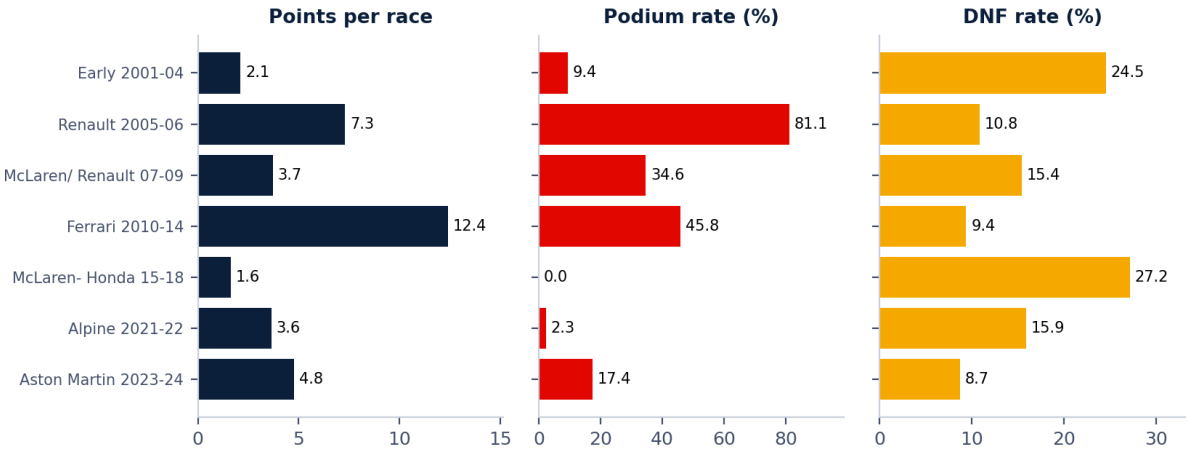


Figure 3. Career phases compared: scoring rate, podium rate and DNF rate.

Phase	Races	Points	Pts/race	Wins	Podiums	DNF %	Avg grid	Avg finish
Early 2001-04	53	112	2.11	1	5	24.5	8.4	6.8
Renault 2005-06	37	270	7.30	14	30	10.8	3.4	2.3
McLaren/ Renault 07-09	52	193	3.71	7	18	15.4	7.5	6.2
Ferrari 2010-14	96	1192	12.42	10	44	9.4	6.3	5.8
McLaren- Honda 15-18	81	133	1.64	0	0	27.2	10.5	9.2
Alpine 2021-22	44	160	3.64	0	1	15.9	10.0	9.0
Aston Martin 2023-24	46	219	4.76	0	8	8.7	9.8	8.9

Finding 2: Team choice explains more than any other factor

Ferrari delivered 1,192 points at 12.4 per race, almost five times McLaren's 2.5 per race even though both teams gave him a similar number of starts (96 and 98). Renault is the team of the two titles: 18 wins, 16 poles and 40 podiums. McLaren is the weakest return on a long stay, with 4 wins and the highest retirement rate among the long spells (24.5%).

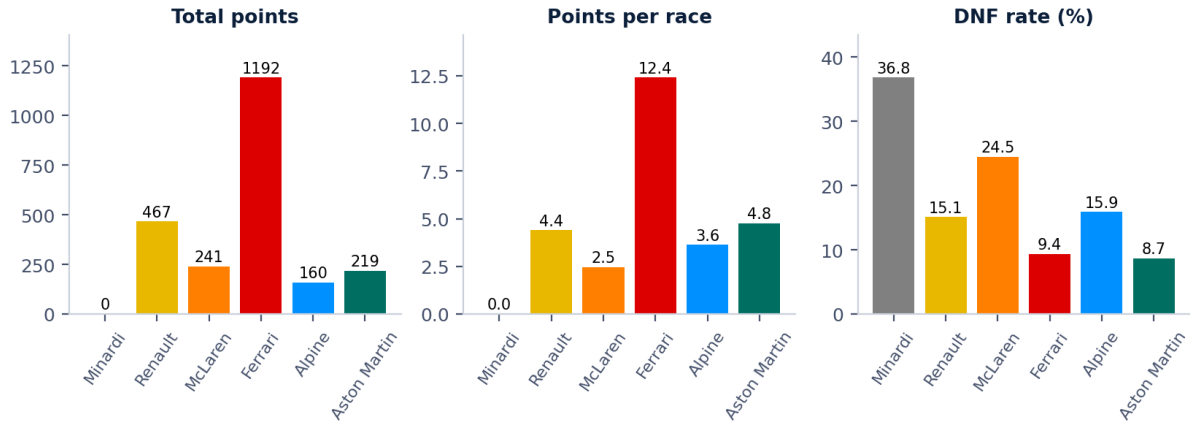


Figure 4. Team comparison: total points, points per race and DNF rate (Minardi 2001 had 36.8% DNF and no points).

Team	Years	Races	Points	Wins	Podiums	Poles	Pts/race	DNF %
Minardi	2001	19	0	0	1	0	0.00	36.8
Renault	2003-09	106	467	18	40	16	4.41	15.1
McLaren	2007, 2015-18	98	241	4	12	2	2.46	24.5
Ferrari	2010-14	96	1192	10	44	4	12.42	9.4
Alpine	2021-22	44	160	0	1	0	3.64	15.9
Aston Martin	2023-24	46	219	0	8	0	4.76	8.7

Note: the dataset's team years show McLaren in 2007 and again from 2015; 2008-09 results are filed under Renault in the data.

Finding 3: Starting position is the strongest predictor of the result

Grid and finish positions correlate at 0.88 for classified races, and each grid place is worth about 0.83 finishing places. The extremes are stark: starting P1 gave a win in 37% of races and a podium in 88%, while starting P6-P10 produced no wins and starts from P11 or lower produced no podiums at all and a 37% DNF rate.

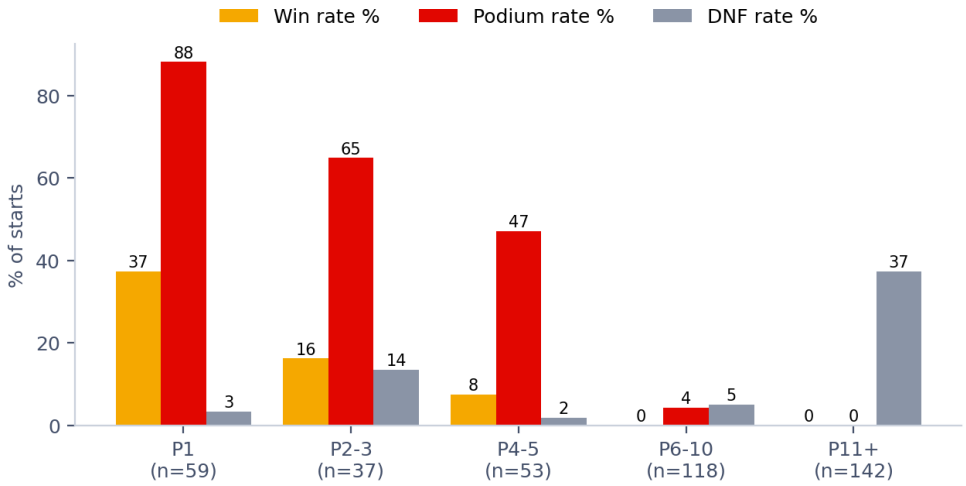


Figure 5. Win, podium and DNF rate by grid slot (P1 means Grid_Position = 1 as recorded).

Grid slot	Starts	Wins	Podiums	Podium %	DNF %	Points/race
P1	59	22	52	88%	3.4%	13.9
P2-P3	37	6	24	65%	13.5%	11.9
P4-P5	53	4	25	47%	1.9%	11.7
P6-P10	118	0	5	4%	5.1%	3.2
P11+	142	0	0	0%	37.3%	0.1

Implication. Every win in his career (32 of 32) started from the first four grid slots. Improving qualifying is the single highest-value lever in this dataset.

Finding 4: Race craft is strong from the middle, but the front-row advantage is not fully converted

Among the 342 classified finishes, he gained 0.16 places on average. From P4-P6 he gained +0.64 and from P11+ he gained +1.01, which points to good overtaking and strategy. The opposite happens at the front: from P1-P3 he lost 1.24 places on average, finishing behind his grid slot in 49 of 89 races. Some of that is structural (a driver on pole cannot gain a place), but the size of the loss, and cases such as 2010 where he started first and finished 16th or 12th, show room to improve first-stint management and strategy.

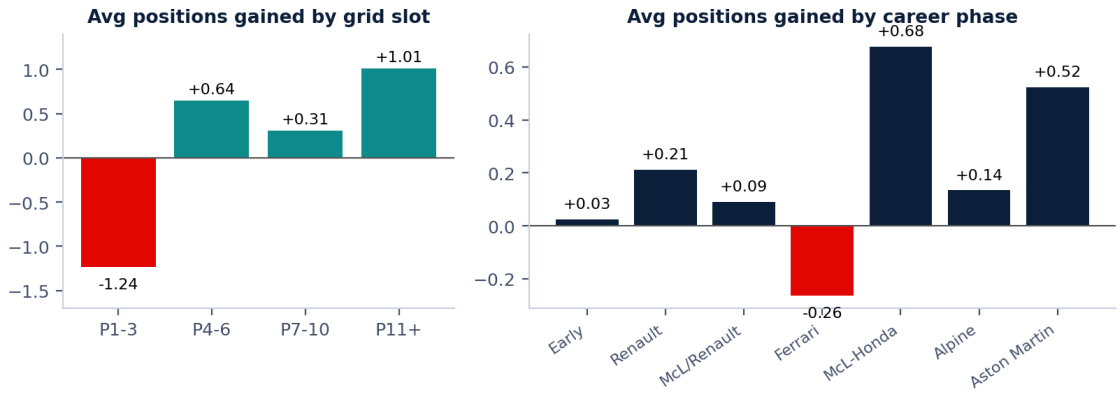


Figure 6. Positions gained or lost: by grid slot (left) and by career phase (right). Retirements excluded.

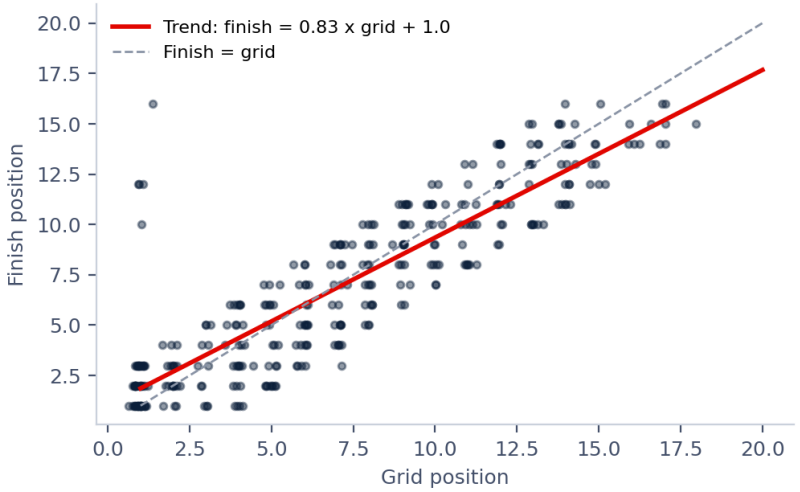


Figure 7. Grid versus finishing position for classified races, with trend line.

Finding 5: Reliability, not pace, cost him the most in difficult years

He retired 67 times. Of these, 46 (69%) were mechanical (hydraulics 13, gearbox 10, engine 9, electrical 7, suspension 7) and 21 (31%) were collisions or accidents (14 and 7). The 2015-18 McLaren-Honda years had 22

retirements in 81 races, 16 of them mechanical, and the 2001 Minardi season saw 7 retirements in 19 races. Using each season's average points from finished races as a proxy, these retirements cost an estimated 300 points, about 13% of his career total. About 123 of those points were lost in the Ferrari years, when the car was fast enough that every retirement hurt.

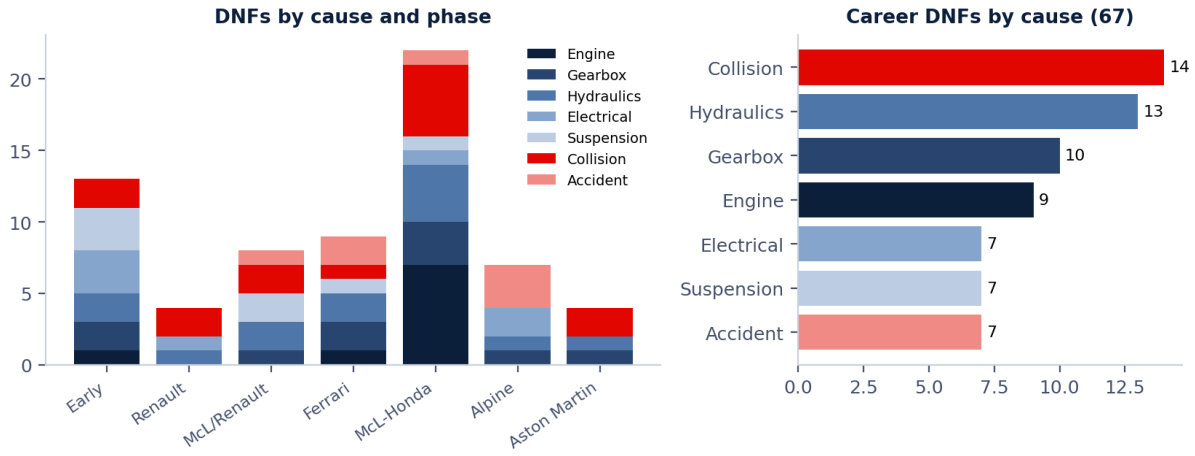


Figure 8. Retirements by cause and career phase (left) and career total by cause (right).

Note: this estimate is a simple proxy, not a reconstruction of what would have happened. Also, 47 of the 67 retirements show the full race distance in Laps_Completed, so the timing of retirements in the data is unreliable and is not analysed here.

Finding 6: Circuit performance varies sharply, and some tracks are consistent weak spots

Points per race range from 9.7 at Barcelona to 1.6 at Yas Marina. The strongest tracks are Barcelona, Losail, Istanbul, Magny-Cours, Montreal (Gilles Villeneuve) and Monaco. The weakest are Yas Marina, Marina Bay, the Red Bull Ring, Bahrain, Sochi and Albert Park. Several weak tracks also carry high retirement rates: Red Bull Ring 4 in 13, Bahrain 4 in 10, Albert Park 4 in 16, Monza 5 in 16, Hockenheim 5 in 11.

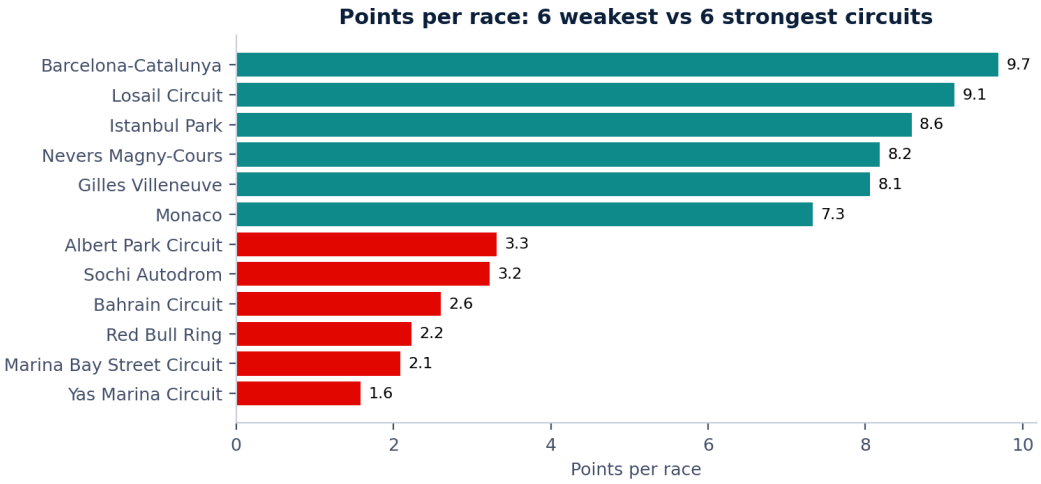


Figure 9. Six strongest and six weakest circuits by points per race (9 to 17 starts per circuit).

By circuit type, permanent tracks are his best (5.8 points per race and 14.4% DNF), street circuits are close (5.4, 17.4%), and **high-speed circuits are weakest (4.9 points per race, 22.1% DNF)** - a pattern that also matches the engine and gearbox failures that dominate his retirements.

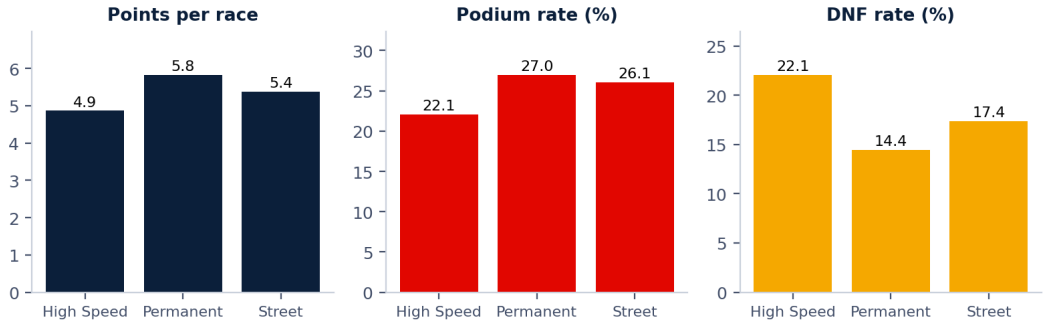


Figure 10. Performance by circuit type.

Finding 7: Recent seasons show a volatile rebound rather than a steady decline

After 80 points in each of 2021 and 2022 at Alpine, 2023 at Aston Martin was a clear step up: 206 points, 8 podiums, average finish 7.8 and just one retirement. The 2024 season fell to 13 points with no podium, an average grid of 10.9 and 3 retirements in 24 races, a points-per-race rate of 0.54 that is lower than every season except his 2001 debut year. Because the swing happened with the same driver, the data points to car competitiveness and qualifying position rather than a decline in racecraft: his average positions gained in 2023-24 (+0.52) is among the best of any phase.

Finding 8: Speed data is not a useful performance explainer here

Average, top and minimum speed are stable across seasons (top speed 335-351 km/h; average 199-221 km/h) and show almost no correlation with points scored (-0.03 to 0.00) or finishing position. This is expected because most values are estimates (Section 8), and because speed differs by circuit far more than by driver. The dashboard's speed line is best read as a context chart for circuit character, not as evidence about his performance.

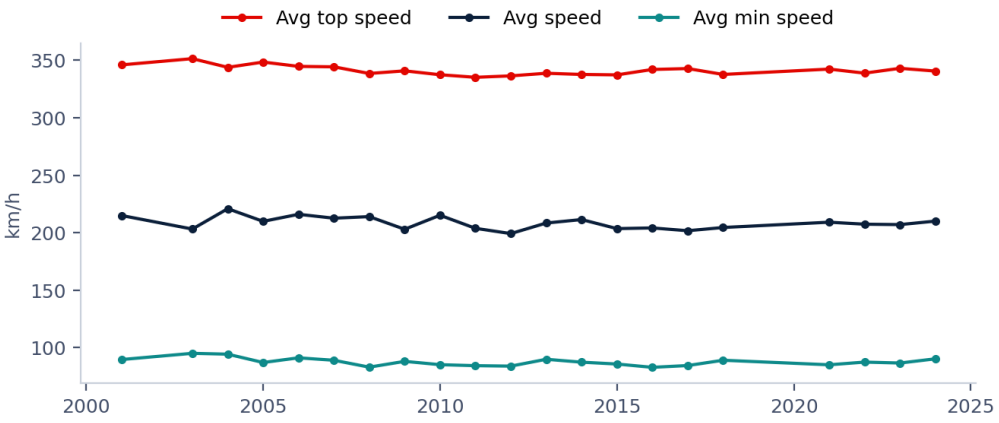


Figure 11. Average, top and minimum speed by season (estimated values).

7. Recommendations: how Alonso can improve performance

Each recommendation links to a specific finding. They describe what the data supports, and should be read as analytical suggestions rather than engineering or coaching instructions.

1

Raise qualifying output first (Finding 3)

Evidence: Grid position explains the result more than any other factor, and all 32 wins came from P1-P4. In 2023-24 his average grid was 9.8, compared with 3.4 in the Renault title years.

Action: Set a qualifying KPI (for example share of races starting in the top 6) and review one-lap pace, tyre preparation and Saturday set-up as a standalone work-stream.

2

Convert front-row starts into wins (Finding 4)

Evidence: From P1-P3 he lost 1.24 places on average and finished behind his grid slot in 55% of classified races. Starting P1 (as recorded) converted to a win only 37% of the time.

Action: Review the first stint and strategy in these races: start procedure, opening-lap defence, tyre degradation and undercut/overcut calls. Track "pole-to-win conversion" as a headline metric.

3

Prioritise reliability when choosing a team and in race management (Finding 5)

Evidence: About 300 points were lost to 67 retirements, 69% of them mechanical. McLaren-Honda had a 27% DNF rate and no podiums.

Action: Use reliability record (DNF rate over prior seasons) as a formal selection criterion for future team moves, and manage components in races where a finish is worth more than a risk.

4

Reduce avoidable first-lap and midfield incidents (Findings 3 and 5)

Evidence: Starts from P11 or lower had a 37% DNF rate versus 2% from P4-P5, and collisions or accidents account for 21 retirements.

Action: When the car qualifies outside the top 10, adopt a lower-risk opening-lap plan: the data shows no podium from that region, so protecting a points finish is worth more than a gamble.

5

Choose the platform, not just the badge (Findings 1 and 2)

Evidence: Points per race: Ferrari 12.4, Renault 4.4 (7.3 in 2005-06), Aston Martin 4.8, Alpine 3.6, McLaren 2.5.

Action: Time team moves around regulation changes and clear evidence of car competitiveness (the 2005 and 2010 jumps show why), and avoid long commitments to projects with weak recent form.

6

Build circuit-specific preparation for weak tracks (Finding 6)

Evidence: Yas Marina (1.6), Marina Bay (2.1), Red Bull Ring (2.2), Bahrain (2.6), Sochi (3.2), Albert Park (3.3) are persistent low scorers, and high-speed layouts show a 22% DNF rate.

Action: Create track dossiers for the six weakest circuits, covering set-up, power-unit loading and tyre plans, and run extra simulator and data review before those rounds.

7

Protect strengths: race craft and top-10 consistency (Findings 3 and 4)

Evidence: He finishes in the top 10 in 63% of races, scores +1.01 places from P11+ and +0.64 from P4-P6, and reached 8 podiums in 2023 from an average grid of 8.5.

Action: Keep the aggressive-but-clean overtaking and strategy approach that makes mid-grid starts productive. The best way to benefit is to feed it with better grid positions (Recommendation 1).

Suggested KPI scorecard to track improvement

Metric	Career baseline	2023-24 (Aston Martin)	Direction
Average grid position	8.1	9.8	Lower is better
Points per race	5.57	4.76	Higher is better
DNF rate	16.4%	8.7%	Lower is better
Top-10 finish rate	62.8%	58.7%	Higher is better
Win rate from grid P1	37%	0 of 1 starts	Higher is better

8. Limitations and data caveats

Good analysis states what it cannot prove. Readers should keep these points in mind before acting on any finding.

- **Speed data is mostly estimated.** 389 of 409 average and fastest-lap values, and every top and minimum speed value, were missing in the raw file and are filled with estimates in the cleaned file. Even where raw and cleaned fastest-lap speeds both exist (20 races), they differ by about 10 km/h on average. Treat the speed visuals as illustrative.
- **Pole flag versus grid position.** Grid_Position = 1 appears in 59 races but only 22 carry the Pole flag. The dashboard's Pole Positions KPI (22) follows the flag, while the grid-slot analysis in this report uses Grid_Position as recorded. The inconsistency should be resolved at source (for example, grid penalties or promoted grid slots) before publishing grid-based percentages.
- **Calendar and circuit chronology.** The race calendar assigns circuits to seasons in ways that do not match real-world history (for example a Miami GP and a Sochi GP in 2001, and Las Vegas appearing across many seasons). Circuit-level findings therefore describe patterns in this dataset and should be treated as indicative, not as historical fact about specific events.
- **Small samples.** Each circuit has only 9 to 17 starts, and each team-era has 2 to 12 seasons. A single bad weekend moves a circuit average noticeably.
- **Estimated points lost.** The 300-point figure is a proxy using season average points from finished races. It assumes a retirement would otherwise have produced an average result.
- **Correlation is not cause.** Grid position and car quality are linked: a better car qualifies higher and finishes higher. The data cannot separate driver and machine contributions without teammate comparisons, which are not in this dataset.
- **Scoring systems changed.** Raw points are not comparable across eras, so points per race and rate metrics are used wherever eras are compared. Points_Scored also does not always map to one value per finishing place within an era (for example a win ranges from 11 to 34 points), which suggests extra scoring rules or inconsistencies in the source.
- **Missing seasons.** 2002, 2019 and 2020 have no race records, which breaks continuity in year-on-year trend lines. The dashboard's Career Span card says 23 seasons, but the data holds 21 (see Section 9.2).

Recommendation for the portfolio. Add a "Data Notes" panel to the dashboard stating that speed values are estimates and listing the pole-flag and calendar caveats. It shows analytical honesty, which recruiters value as much as polish.

9. Conclusion and next steps

Starting from seven partly inconsistent CSV files, the project produced a cleaned, validated data model and a Power BI dashboard that reconciles to Alonso's recognised career totals: 32 wins, 106 podiums, 22 poles and 2 titles across 409 starts. The analysis shows that his results follow the competitiveness of the car and his qualifying position more than anything else, that front-row starts are not always converted, and that retirements, mostly mechanical, cost an estimated 300 points.

The practical message is consistent: **qualify higher, convert front-row starts, and avoid retirements**.

9.1 Suggested improvements to the dashboard

- Add a **Career Phase** slicer (Renault peak, Ferrari, McLaren-Honda, and so on), since phases explain more than individual seasons.
- Add a **Grid vs Finish** scatter and a **Positions Gained by Grid Slot** bar to expose the front-row leak (Findings 3 and 4).
- Add a **DNF by Cause** stacked bar, split mechanical versus collision, to make reliability visible beyond the gauge.
- Add a **Data Notes** panel flagging estimated speed values and the pole-flag caveat.
- Add teammate comparison data to separate driver from car performance.

9.2 Dashboard-to-report reconciliation

The published dashboard screenshot was compared with the figures in this report. All numeric visuals match; three label issues are worth fixing.

Dashboard item	Dashboard shows	Report / data	Status
KPI cards	409 races, 67 DNF, 106 podiums, 22 poles, 26 fastest laps, 32 wins, 2 titles, 5.57 pts/race, 341.21 top speed, 2K points	Identical (2K = 2,279 rounded)	Match
DNF gauge	16.38%	16.4%	Match
Finishing Position Breakdown	32 / 37 / 37 / 151 / 152	32 / 37 / 37 / 151 / 152	Match
Team charts	Ferrari 1.2K, Renault 0.5K pts; wins 18 / 10 / 4; races 106, 98, 96, 46, 44, 19	Identical	Match
Circuit type	Permanent 5.8, Street 5.4, High Speed 4.9	5.83, 5.38, 4.87	Match
Season table and trend chart	2012: 279 pts, 14 podiums; 2023: 206 pts, 8 podiums	Identical	Match
Career Span card	23 Seasons	21 seasons in the data	Fix label
Timeline: 2012	"Returned to McLaren"	Milestones file: 2012 = Runner-Up (Ferrari); return to McLaren = 2015	Fix label
Timeline: 2021 and 2023	2021 "Returned to F1 (Aston Martin)"	Milestones file: 2021 return was with Alpine; Aston Martin from 2023	Fix label

Career Span appears to be calculated as end year minus start year (2024 - 2001 = 23); a distinct count of seasons gives 21. The timeline text looks hard-coded rather than driven by the Career Milestones table, which is why it can drift from the data.

Appendix A. Season-by-season performance

Season	Team	Races	Points	Wins	Podiums	Poles	DNF	Avg grid	Avg finish
2001	Minardi	19	0	0	1	0	7	9.0	6.4
2003	Renault	16	55	1	1	1	3	8.1	7.0
2004	Renault	18	57	0	3	0	3	8.0	6.9
2005	Renault	19	132	7	16	7	2	3.6	2.1
2006	Renault	18	138	7	14	7	2	3.2	2.6
2007	McLaren	17	108	4	12	2	2	3.8	3.1
2008	Renault	18	60	2	4	1	3	8.9	7.5
2009	Renault	17	25	1	2	0	3	9.6	8.0
2010	Ferrari	19	254	5	11	3	1	5.1	5.9
2011	Ferrari	19	256	1	11	0	2	5.3	4.3
2012	Ferrari	20	279	2	14	1	2	3.5	3.6
2013	Ferrari	19	242	2	6	0	2	6.7	5.9
2014	Ferrari	19	161	0	2	0	2	10.9	9.2
2015	McLaren	19	11	0	0	0	7	11.8	9.8
2016	McLaren	21	54	0	0	0	6	9.3	8.4
2017	McLaren	20	18	0	0	0	5	11.5	8.3
2018	McLaren	21	50	0	0	0	4	9.7	10.2
2021	Alpine	22	80	0	1	0	4	9.4	8.6
2022	Alpine	22	80	0	0	0	3	10.7	9.5
2023	Aston Martin	22	206	0	8	0	1	8.5	7.8
2024	Aston Martin	24	13	0	0	0	3	10.9	10.0
Total		**409**	**2279**	**32**	**106**	**22**	**67**	**8.1**	**6.9**

Average finish excludes retirements (no classified position). Source: 03_Race_Results_Clean.csv.

Appendix B. Dashboard measures

Measure	Definition used (logic)
Total Races	Count of rows in Race Results
Wins	Sum of Win_Flag
Podiums	Sum of Podium_Flag
Pole Positions	Sum of Pole_Flag
Fastest Laps	Sum of Fastest_Lap_Flag
DNF	Sum of DNF_Flag
DNF Rate	DNF divided by Total Races
Total Points	Sum of Points_Scored
Points Per Race	Total Points divided by Total Races
Championships	From Driver Profile (2)
Avg Speed / Avg Top Speed / Avg Min Speed	Averages of the Speed Analysis columns (estimated values)

The measure names come from the .pbix file. The logic column describes the standard definition of each measure, because the compressed DAX expressions are not readable outside Power BI.

Appendix C. File inventory

File	Type	Notes
Fernando_Alonso.pbix	Power BI report	1 page, 32 visual containers
01-07 *.csv	Raw data	Original exports with the issues listed in Section 3
01-07 *_Clean.csv	Cleaned data	Same rows, corrected status/flags, added helper columns, filled speed estimates